The Rock Cycle

After the Program

1. Start a rock collection. Collect rocks from various locations and try to identify them. Try to find at least three different rocks and draw a picture of each one. Remember that rocks come in many different sizes. Select rocks that are small enough to pick up and examine with the hand. Answer the following questions about your rock:

   a) Where did you find this rock?

   b) How would you describe this rock?

   c) What does it feel like? Are parts of it smooth? Are parts of it rough?

   d) How is it different from the other rocks you found?

   e) Do you see any patterns in this rock? Look at the whole rock. Does it look like it is all made of the same thing or does it look like it is made up of smaller parts?

2. What is the difference between a mineral and a rock?

3. Different rocks have different uses depending on their properties. Select a rock and investigate it. Include type, properties and its uses. Why is your rock suited to its particular use? For a list of rocks to choose from go to: http://en.wikipedia.org/wiki/List_of_rocks

4. Geologists use many different methods to identify minerals that make up rocks. Each mineral possesses a unique set of identifying properties. Select three minerals and list the identifying properties each one has.

5. Around 90 minerals are classed as gems, but only about 20 of these are important gems. Gemstones are often what people mean when they talk about "crystals". There are many gemstones and most are used for jewellery or decoration. Some gemstones look similar to what the mineral looks like when found in nature and others are very different.

   - Activity: Take A Gemstone Safari. Go to http://www.africagems.com/gemsafari.html. This site sells gemstones from Africa. It has photos of coloured gemstone mining in Africa and has an article about processing gemstones. Select a gemstone and create a poster that includes the gem's value, a description, properties, availability and mining techniques.

6. Most people are familiar with precious and semi-precious gemstones because of the popularity of birthstones. Select your birthstone from the chart below and find out information about it such as how and where it is formed, mined and processed.

Month	Modern (as of 1912 USA)	Traditional
January	Garnet	Garnet
February	Amethyst	Amethyst
March	Aquamarine	Bloodstone
April	Diamond	Diamond
May	Emerald	Emerald
June	Pearl, Moonstone	Alexandrite
July	Ruby	Ruby
August	Peridot	Sardonyx
September	Sapphire	Sapphire
October	Opal, Tourmaline	Tourmaline
November	Yellow Topaz, Citrine	Citrine
December	Blue Topaz, Turquoise	Zircon, Turquoise, Lapis Lazuli

7. Rocks can contain crystals of various shapes. The shapes (crystalline structures) are ordered arrangements which are unique to a particular crystal. Look up various crystals and draw their shapes. Where would you typically find these crystals?